



Modelling and Analysis of Pouch Lithium-Ion Cells in Electric Vehicle Combined Single and Dual Battery Packs Integrated With and Without Active Cooling

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Abstract: This study presents detailed modelling and comparative analysis of pouch lithium-ion cells (PLCs) configured in combined single and dual battery packs for electric vehicles (EVs), appraising their performance with and without active cooling systems. Pouch cells were selected due to their high volumetric energy density (VED), lightweight design, and suitability for high-power EV applications. Using SolidWorks for computer aided design (CAD) modelling, Analysis System (ANSYS) Fluent software with the multi-scale multi-domain (MSMD) approach, and equivalent circuit model (ECM), simulation based thermal and electrochemical behaviour during 1-hour charging process at specified currents was achieved. Four configurations were examined: single-pack and dual-pack, each with and without liquid active cooling. Main parameters included state of charge (SoC), temperature distribution, and thermal gradients. Results confirmed that dual-pack configuration significantly outperformed single-pack, achieving over 69.3% higher SoC (55.07% against 32.53%) under the same conditions. Without active cooling, maximum temperatures reached 541.11 K (single) and 462.77 K (dual), indicating notable hotspots in the single pack. Active cooling dramatically reduced temperatures to 295 K across both setups, with the dual pack displaying superior uniformity and a 0.45% lower average temperature, adequately averting thermal gradients and enhancing safety. Mathematical validation of SoC dynamics confirmed the dual configuration's theoretical advantage in charging efficiency, tempered by heat losses. Findings reveal that dual battery packs integrated with active cooling render optimal equilibrium of faster charging, improved thermal management, and extended battery life, addressing limitations associated with conventional single-pack designs.

Keywords: Electric vehicle; Pouch lithium-ion cells; Active cooling; Single and dual battery packs

1 Introduction

Most conventional electric vehicles (EVs) are driven by lithium-ion (Li-ion) batteries, owing to their cost efficiency, high energy density and power capability. Amidst the available Li-ion models, pouch cells tend to be more outstanding for EV operations. They come in a soft aluminum-polymer laminate sheath that allows minimized weight, adequate packing as well as superior volumetric energy density (VED) in comparison with prismatic or cylindrical cells. Pouch cells utilize a number of chemistries such as Nickel Manganese Cobalt (NMC) or Nickel Cobalt Aluminium (NCA) with graphite anodes (high energy density chemistries domicile in EV pouch batteries), making them appropriate for high-energy-high-power requirements in EVs [1, 2]. In recent times, a number of relevant studies have been carried out in this field. For example, Koniak and Czerepicki [3] identified temperature, operation current and the level of discharge as some of the factors that influence the performance of EV batteries. Consequently, extensive evaluation of some factors including route characteristics, volume and weight, charging currents, discharge level, as well as seasonal operating temperatures was suggested. The study analyzed various technical solutions via appraisal of the weighting sum of specific factors for each choice considered. It was concluded that proper battery selection and adequate performance can prolong the service life between replacements, thereby,

plummeting operating costs of the electric power-driven vehicle. Lu et al. [4] engaged a refined electric-thermoaging coupled model for every single cell in a battery pack to digest external and internal variations. This technic was confirmed on a corporate battery pack configured in three-parallel six-series (3P6S), indicating an outstanding upsurge in charged rate of 39.2% in 10 minutes and 92.2% in 53 minutes at 25 °C, exceeding earlier charging procedures. A 30.7 kWh battery pack of 48 high-nickel Li-ion cells, with typical volumetric heat emission of 186,360 W/m³ through 3C charging was examined by Jiang et al. [5]. For flow rate of 10 L min⁻¹ as well as original ambient temperature of 40 °C, peak temperature of 55.4 °C was attended by the battery pack in the course of 3C fast charging which was within the limit of tolerance. On the other hand, 46.48% enhancement in heating rate (1.04 °C·min⁻¹) was achieved via top-inlet/bottom-outlet flow geometry compared to traditional bottom-inlet geometry which minimized the peak intra-pack temperature variations to 3.9 °C, gaining over 59.38% reduction at after heating. To estimate the battery temperature of EVs, Hong et al. [6] employed a multi-parameter disparity-grounded vehicle identification grouping approach to cluster the charging data by means of vehicle identification. Afterwards, Long Short-Term Memory (LSTM) neural networks based on actual accumulated charging data were employed to estimate the battery temperature based on the total and clustered data. The outcome indicated that, the correctness of the approach of identification followed by prediction is optimum, as the mean relative error for the peak estimated temperature declined from 1.45–1.06%. The effect of different electrical configurations, tab width, tab depth, busbar height as well as discharge rate on the electrical and thermal efficiency of LiMn₂O₄ battery cell pack was examined by Mishra et al. [7]. From the voltage and power characteristics assessment, it was observed that expanding the totality of parallel connections resulted in reduction of the entire voltage and power output while considerably improving discharge time, discharge rate impacts ($T_{\max}$ by 44% and $\Delta T_{\max}$ by 58.2%), and electrical configuration (42.5% and 38.6%). This implies that appropriate stability in discharge rate, tab dimensions, busbar height, and electrical configuration is essential in the process of designing Li-ion battery packs for EVs. However, the thin or flat geometry of pouch cells exhibit a certain issue that can cause in-service thermal gradients, particularly under high C-rates, typically, fast charging/discharging or peak power occurrences such as acceleration in urban air mobility. The generation of heat originates from entropic changes, Joule heating (ohmic losses), and operation reactions. On the other hand, extreme temperatures stimulate degradation, limit lifespan, and compromise performance, necessitating effective thermal management [8–10]. From a thermal-engineering standpoint, pack layout is central to this problem: in the single-pack arrangement, all six cells and their heat-generating tabs are consolidated within one continuous, tightly packed stack, so heat accumulates faster than it can conduct outward, producing pronounced hotspots at the centrally located cells and a non-uniform temperature field across the pack. Splitting the same number of cells into two smaller, physically separated three-cell packs increases the exposed surface-area-to-volume ratio of each sub-pack and shortens the internal conduction path from the cell core to the boundary, allowing heat to be dissipated more readily; this lowers peak hotspot intensity and yields a more uniform temperature distribution across the dual-pack arrangement, forming the central heat-transfer motivation for comparing the two layouts in this study. From related literature, it has been noted that thermal management is a crucial aspect in setting up an efficient EVs (since it drives the efficiency and longevity of the batteries), and the several solutions (liquid cooling systems and phase change materials (PCMs) among other conventional techniques) established to enhance heat dissipation [11, 12]. It has been further observed that some of the key issues that can potentially hamper the efficiency of these solutions are the complexity of the systems, costs implication, and their integration with the vehicle components [13, 14]. Findings from related literature suggested advanced materials, thermal control algorithms, and integrated design techniques that addresses efficiency. However, proposed research emphasis by Gómez Díaz et al. [15] and Das et al. [16] included enhancing the effectiveness of thermal models, designing efficient and viable thermal management systems, and considering emerging technologies (immersion cooling and nanotechnology). In the absence of active cooling, Li-ion batteries primarily rely on natural convection, radiation, phase change materials, or heat pipes for heat dissipation. Although these passive techniques are energy-efficient and structurally simple, their heat-removal capability can become inadequate under high-load operation, particularly for large-format pouch cells where non-uniform heat generation can produce significant spatial temperature gradients and consequently accelerate non-uniform ageing [17, 18]. However, the integration of active cooling systems utilize forced air, liquid (such as water-glycol openings or cold plates), or hybrid techniques. For Air cooling integrated pouch cells, these systems are lightweight and cost efficient with inadequate cooling capacity effective at ≥ 4 m/s velocities for modest loads [19], compared to liquid cooling which offers a more effective cooling outcome. It comes with different configurations such as two-side cooling, bottom cooling as well as edge cooling designed with unique channels typically streamlined or honeycomb fins. The two-sided liquid cooling configuration is known for optimal performance that outweigh the single-sided or ambient setups (particularly at high C-rates) in terms of reduction of high operating temperatures [20, 21]. A study comparing 1P4S configurations (with 1 parallel and 4 series) pouch module under variable ambient temperature conditions, coolant flows and discharge rates, indicated that active systems sustain cells within optimal range 20–40 °C and $\Delta T < 5$ °C for improved safety, efficiency and longevity [22]. Recent research on dual-pack battery configurations reveal that adequately designed multi-pack architectures, integrated with appropriate battery thermal-management

systems can improve thermal performance, operational flexibility, as well as system-level resilience compared to conventional single large packs, even though the introduction of additional structural and control complexity may be involved. On this basis, cutting-edge electrochemical-thermal and multiphysics simulations have used progressively to typify battery thermal performance and predict thermal-runaway initiation and proliferation in emerging battery-pack designs [23–25]. This plays a vital role in optimizing coolant flow uniformity, preload forces (to manage swelling), and material interfaces. Generally, conventional modelling and simulation of pouch cells in single and dual EV battery packs, with integrated or non-integrated active cooling has initiated the novelties in in in battery thermal management systems (BTMS) [26, 27]. This approach has equated performance, safety, energy efficiency, and predictability, sustaining the evolution to higher-performance, longer-range, and safer EVs. This forms the basis for conventional studies which consistently lay emphasis on sustainable materials, hybrid cooling, and Artificial Intelligence (AI)-optimized controls [28–30]. The present study seeks to model and analyze a combined single and dual battery packs integrated with and without active cooling, with the battery cells represented as pouch lithium-ion cells (PLCs) rather than modelling a full-scale battery with intricate details.

2 Research Methodology

The design phase of the simulation was carried out using SolidWorks to model the battery configuration. Given the goal of reducing computational resource demand, the complexity of a large battery pack (which typically contains hundreds of individual cells) was simplified. Instead of modelling a full-scale battery with intricate details, the battery cells were represented as PLCs, a widely used battery format in EVs. These cells feature a rectangular active region with two square terminals or “tabs” for the positive and negative connections. The single configuration consisted of six cells connected in series using five separate bussbars, while the dual configuration consisted of three cells each also connected in series by two bussbars. The final designs were saved in acis format for compatibility with Analysis System (ANSYS). Computer aided design (CAD) model of single configuration without and with active cooling and CAD model of dual configurations without and with active cooling are presented in Figure 1.

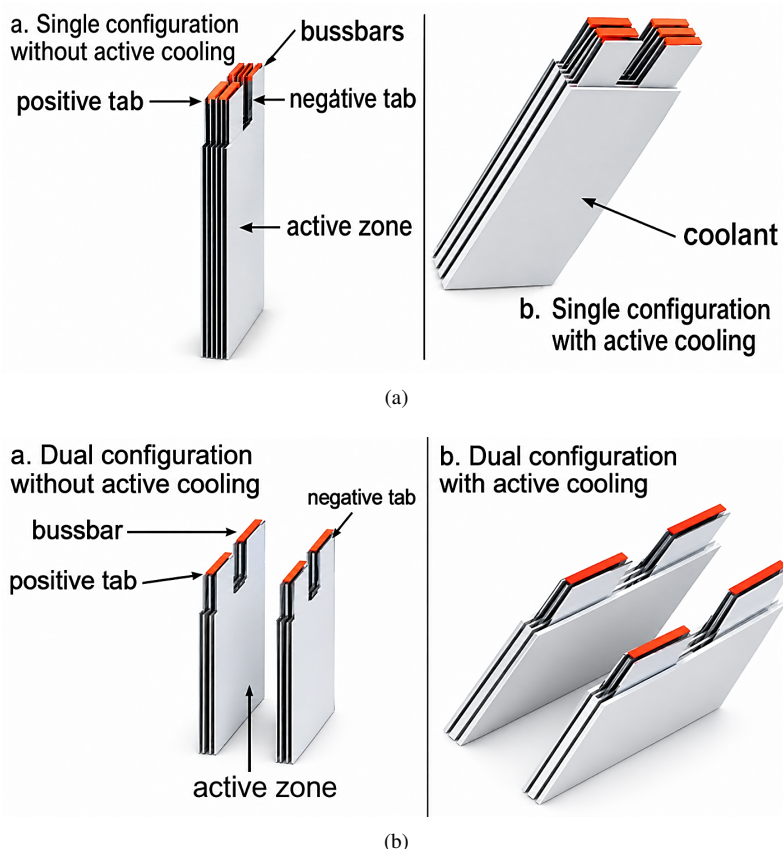


Figure 1. Computer aided design (CAD) model of: (a) single configuration without and with active cooling; (b) dual configurations without and with active cooling

Figure 2 represents a cross sectional view of Li-ion battery system designed for EVs, incorporating numerous pouch cells arranged in a stacked configuration layout. The major components include positive and negative tabs for electrical connections, copper bussbars on top for current assortment, as well as adequate cooling channel located

along the base for active cooling effects. This layout is mostly suited for dual-pack active cooling configuration, which offers effective extraction of heat from the cell. Structural firmness and safety are rendered by the module casing while the enter assembly is sealed by the cover.

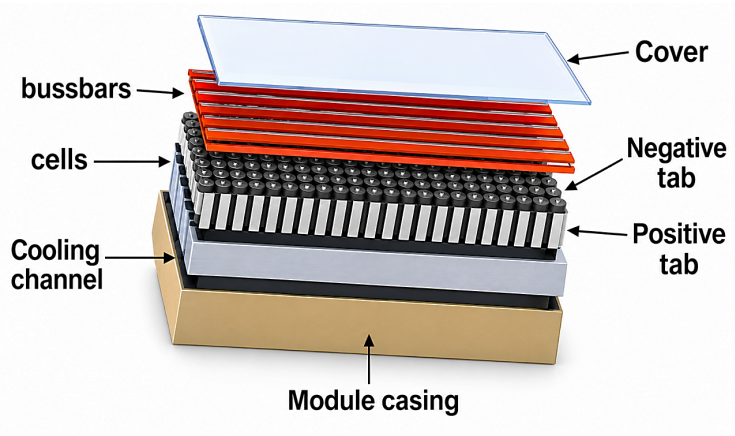


Figure 2. Pouch lithium-ion (Li-ion) battery module with assimilated cooling network for active thermal management in electric vehicle (EV) packs

The CAD models were then imported into ANSYS Fluent using the Fluid Flow (Fluent) analysis system, where the next stage of the simulation began. In Fluent, a meshing process was applied to discretize the model. Care was taken to ensure a high-quality mesh that would capture the key thermal and electrical characteristics of the system. Several named selections were defined to facilitate the setup, including labels for the active region, positive tab, negative tab, and busbars. Mesh details of the single and dual pack with and without active cooling are presented in Table 1, while the mesh visualization is illustrated in Figure 3.

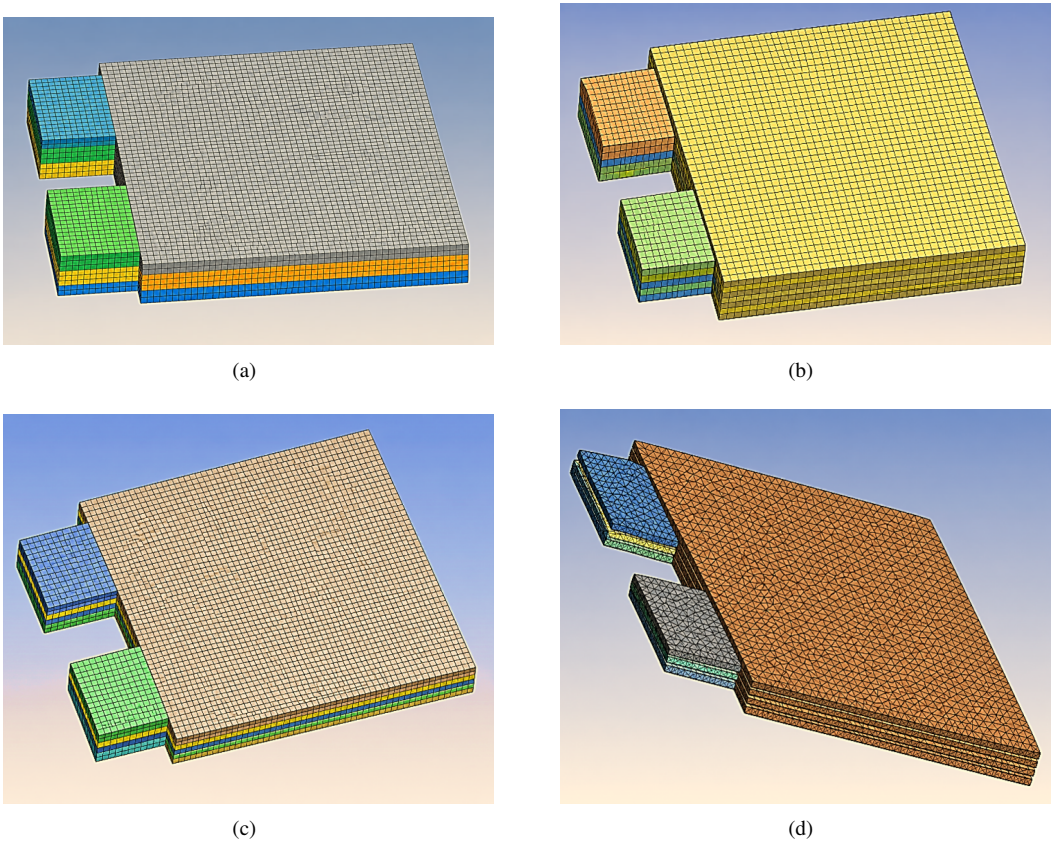


Figure 3. Mesh model for: (a) single pack without active cooling; (b) single pack with active cooling; (c) dual pack without active cooling; (d) dual pack with active cooling

Table 1. Mesh details of the single and dual pack with and without active cooling

Single Pack Without Active Cooling			Single Pack With Active Cooling		
Cells	Faces	Nodes	Cells	Faces	Nodes
48,741	197,731	100,920	44,066	327,189	90,864
Dual Pack Without Active Cooling			Dual Pack With Active Cooling		
Cells	Faces	Nodes	Cells	Faces	Nodes
24,336	98,680	50,364	53,424	384,199	109,340

After meshing, the model was transferred to ANSYS Fluent for detailed simulation setup. The battery model was activated, along with the energy model physics (Table 2) to account for heat generation and transfer during charging. The solution method was set to multi-scale multi-domain (MSMD), a commonly used approach for battery modelling and simulations, which allows for accurate prediction of the electrochemical and thermal behavior of the cells [31]. Additionally, the equivalent circuit model (ECM) was enabled to simulate the behavior of the Li-ion cells during the charging process. Conductive zones and electric contacts were assigned to model the flow of current and heat within the system, ensuring proper representation of electrical interactions. Parameters specific to the ECM were also defined to represent the internal behavior of the battery cells. These settings allowed Fluent to calculate the heat generation, internal resistance, and state of charge (SoC) dynamics during charging. The material properties of the battery components (Table 3), such as thermal conductivity and electrical resistivity, were defined for the active zone, tabs, and busbars.

Table 2. Energy model physics for single pack without and with active cooling

Single Pack With and Without Active Cooling		Dual Pack With and Without Active Cooling	
Model	Settings	Model	Settings
Space	3D	Space	3D
Time	Unsteady, 1st-order implicit	Time	Unsteady, 1st-order implicit
Viscous	SST k - ω turbulence model	Viscous	SST k - ω turbulence model
Heat transfer	Enabled	Heat transfer	Enabled
Battery model	Enabled	Battery model	Enabled
Model Options	Settings	Model Options	Settings
Solution method	MSMD	Solution method	MSMD
E-chem model	ECM	E-chem model	ECM
Nominal cell capacity	214 Ah	Nominal cell capacity	107 Ah
Specified system current	-48 A	Specified system current	-48 A
E-chem stop criterion	SoC-min and max: 1	E-chem stop criterion	SoC-min and max: 1
Model Parameters	Settings	Model Parameters	Settings
Initial SoC	0.1 (10%)	Initial SoC	0.1 (10%)
Reference capacity	214 Ah	Reference capacity	107 Ah

Note: 3D = three-dimensional; SST = shear-stress transport; MSMD = multi-scale multi-domain; ECM = equivalent circuit model; SoC = state of charge; E-chem = electrochemistry.

The cell zones were set to capture internal heat generation, while the boundary conditions (Table 4) ensured accurate heat transfer and electrical flow at the interfaces between the tabs and busbars. Once all parameters were configured, report plots were created to track two critical metrics over time: SoC and temperature. Before running the simulation, a hybrid initialization was performed to set initial conditions across the model, ensuring the simulation would converge properly. The simulation was run for a 1-hour charging period, tracking the evolution of both SoC and temperature throughout the process.

For clarity, the active-cooling cases use water (properties in Table 3) as the sole cooling medium; no forced-air cooling was modelled anywhere in this study. The coolant channel runs along the base of the pack beneath the cell stack, as shown in Figure 2, entering at a uniform velocity of 0.5 m/s and an inlet temperature of 290 K (Table 4). The channel outlet was set as a pressure outlet at atmospheric gauge pressure, allowing the coolant to exit freely as it absorbs heat along the base of the pack. SolidWorks CAD model of an EV rolling chassis designed to study the in-service performance of PLCs layout stacked in single and dual battery pack configurations, as shown in Figure 4. The chassis assembly is presented briefly to illustrate the in-vehicle packaging context of the two pack layouts; a detailed structural analysis of the chassis, suspension and drivetrain components is outside the scope of the present heat-transfer study. The sectional battery layout instead enables evaluation of cooling efficiency on the PLCs during

charging and discharging. The simulation was conducted for both the single and dual battery configurations, with and without active cooling. The results were computed, and contour plots for SoC and temperature distribution were generated for each configuration. The plots provided detailed insights into how charge and heat were distributed within the cells over time. The results for each configuration were compared, pinpointing differences in charging performance and thermal management between the single and dual battery setups, with specific attention to the role of active cooling in improving thermal regulation. These findings were used to draw conclusions regarding the potential benefits of the dual-battery configuration for reducing charging time and managing thermal effects more effectively.

Table 3. Material properties of the battery components

Single and Dual Pack Without Active Cooling		Single and Dual Pack With Active Cooling	
Fluid (Air)		Fluid (Air)	
Density	1.225 kg/m ³	Density	1.225 kg/m ³
Cp (specific heat)	1,006.43 J/(kg·K)	Cp (specific heat)	1,006.43 J/(kg·K)
Thermal conductivity	0.0242 W/(m·K)	Thermal conductivity	0.0242 W/(m·K)
Viscosity	1.7894×10^{-5} kg/(m·s)	Viscosity	1.7894×10^{-5} kg/(m·s)
Molecular weight	28.966 kg/kmol	Molecular weight	28.966 kg/kmol
Solid (Tab Material)		Water	
Density	8,978 kg/m ³	Density	998.2 kg/m ³
Cp (specific heat)	381 J/(kg·K)	Cp (specific heat)	4,182 J/(kg·K)
Thermal conductivity	387.6 W/(m·K)	Thermal conductivity	0.6 W/(m·K)
UDS diffusivity	10,000,000 kg/(m·s)	Viscosity	0.001003 kg/(m·s)
-	-	Molecular weight	18.0152 kg/kmol
Active Zone Material		Solid (Tab Material)	
Density	2,092 kg/m ³	Density	8,978 kg/m ³
Cp (specific heat)	871 J/(kg·K)	Cp (specific heat)	381 J/(kg·K)
Thermal conductivity	202.4 W/(m·K)	Thermal conductivity	387.6 W/(m·K)
UDS diffusivity	35,410,000 kg/(m·s)	UDS diffusivity	10,000,000 kg/(m·s)
Active Zone Material		Active Zone Material	
-	-	Density	2,092 kg/m ³
-	-	Cp (specific heat)	871 J/(kg·K)
-	-	Thermal conductivity	202.4 W/(m·K)
-	-	UDS diffusivity	35,410,000 kg/(m·s)

Note: Cp = capacity; UDS = user-defined scalar; (-) = not applicable.

Table 4. Boundary condition for single and dual pack with and without active cooling

Single and Dual Pack Without Active Cooling		Single and Dual Pack With Active Cooling	
Type	Thermal	Type	Thermal
Condition	Convection	Condition	Convection
Heat transfer coefficient	25 W/(m ² ·K)	Heat transfer coefficient	25 W/(m ² ·K)
Free stream temperature	300 K	Free stream temperature	300 K
-	-	Coolant velocity	0.5 m/s
-	-	Coolant initial temperature	290 K

Note: (-) = not applicable.

3 Results and Discussion

The simulations were conducted with a focus on key parameters such as SoC and temperature distribution during a 1-hour charging period. The performance of the battery systems was evaluated with and without active cooling, providing insights into the thermal behavior and charging efficiency of the configurations. By comparing the results, this discussion aims to assess the potential benefits of the dual-battery configuration in reducing charging time and enhancing thermal management. Additionally, the effectiveness of active cooling in mitigating temperature rise during charging will be evaluated. The findings are analyzed to draw conclusions regarding the optimal configuration for improving EV battery charging performance and safety. The simulation results (Table 5) provide

a detailed comparison of the performance of single and dual battery configurations in terms of SoC and average temperature, both with and without active cooling. Maximum battery temperature of 541.11 K was recorded as the highest for single pack without active cooling, followed by 462.7719 K for dual pack without active cooling. However, maximum temperature of 295.6208 and 294.1847 K were observed for single and dual pack configurations with active cooling. The uncooled peak values of 541.11 K and 462.77 K are well outside the normal operating range of Li-ion cells and are not put forward as physically realistic in-service temperatures; rather, they arise from the combination of a high, sustained specified current (-48 A) applied to the reduced-scale pack model, a relatively weak natural-convection-only heat-loss boundary ($h = W/(m^2 \cdot K)$, no radiation) at the free surfaces without active cooling, and the simplification inherent in representing the pack with a small number of pouch cells rather than a full-scale, thermally massive assembly. Under real operating conditions such temperatures would be well within the thermal-runaway-onset regime for NMC/NCA pouch chemistries, so these values are considered as an upper-bound, worst-case indicator of the heat-accumulation and hotspot risk inherent in the single-pack layout rather than as a benign or acceptable outcome, indicating why active cooling is treated as essential rather than optional in this study. Analysis of the results with and without active cooling is graphically illustrated in Figure 5.

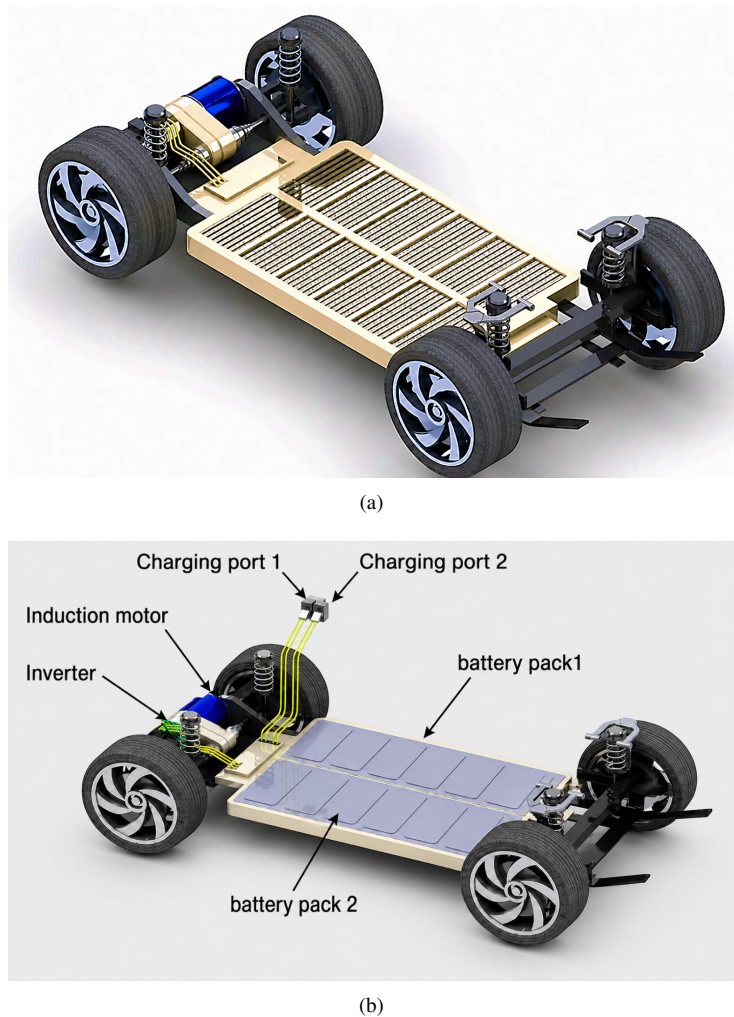


Figure 4. (a) SolidWorks model view of an electric vehicle (EV) base and chassis assembly; (b) model view with battery pack configuration

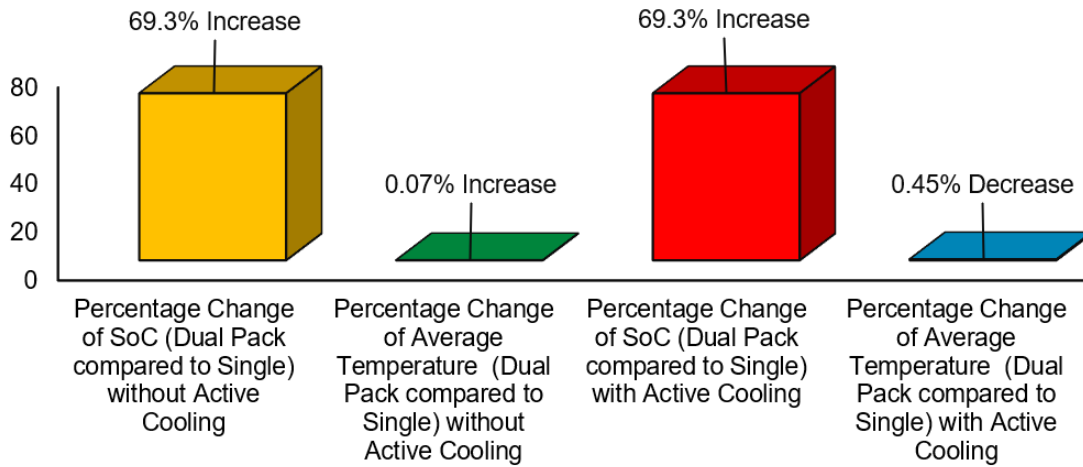
3.1 State of Charge Performance

The SoC indicates the efficiency of the charging process. Without active cooling, the dual battery configuration achieved a SoC of 55.07%, whereas the single pack reached only 32.53%. This represents a 69.3% increase in SoC for the dual battery setup, showing a significant improvement in charging speed compared to the single pack. With active cooling, the SoC remained unchanged for both configurations, with the dual pack still at 55.07% and the single pack at 32.53%. This suggests that the introduction of cooling does not impact the charging efficiency of the battery systems, as both configurations maintained the same charging performance.

Table 5. Simulation results of performance for single and dual pack with and without active cooling

Parameters	Single Pack Without Active Cooling	Dual Pack Without Active Cooling	Single Pack With Active Cooling	Dual Pack With Active Cooling
Temperature vs. time	541.1166 K	462.5001 K	292.9649 K	291.7718 K
Cell-voltage vs. time	11.96532 V	12.03017 V	12.9712 V	12.03006 V
SoC vs. time	0.3253013 (32.53%)	0.5507243 (55.07%)	0.3253013 (32.53%)	0.5507243 (55.07%)
Battery-max-temperature recorded	562.8683 K	462.7719 K	295.6208 K	294.1847 K
Battery-terminal-voltage	71.8293 V	36.10904 V	71.82896 V	36.10708 V
Battery-terminal-current	-48 A	-48 A	-48 A	-48 A
Delta-time	60 s	60 s	60 s	60 s
Iters-per-time step	20	20	20	20
Flow-time	3600 s	3600 s	3600 s	3600 s

Note: SoC = state of charge.

**Figure 5.** Analysis of results with and without active cooling

3.2 Thermal Performance

Thermal management is critical in maintaining battery performance and safety [32]. Without active cooling, the average temperature of the dual battery configuration was at 462.5001 K, compared to 541.1166 K for the single pack without active cooling, reflecting a 0.07% increase. This minor rise in temperature indicates that the faster-charging dual battery setup does not significantly exacerbate thermal conditions. When active cooling was applied, both configurations experienced a considerable drop in temperature. The single battery pack reached an average temperature of 292.9649 K, while the dual pack dropped to 291.7718 K, showing a 0.45% decrease in temperature for the dual battery configuration with active cooling compared to the single with active cooling. This highlights the effectiveness of active cooling in managing heat, particularly in the dual configuration, which showed a slight but notable improvement in thermal performance. To make the thermal comparison more informative, three further quantities were extracted from the same dataset: the temperature reduction achieved by active cooling, the temperature-rise rate during charging, and the intra-pack temperature spread (taken as the difference between the reported maximum and average temperatures). Active cooling reduced the peak battery temperature by 267.25 K (47.5%) in the single pack (562.87 K to 295.62 K) and by 168.59 K (36.4%) in the dual pack (462.77 K to 294.18 K), showing that cooling delivers a proportionally greater benefit to the single-pack layout, which starts from the more severe thermal condition. Over the 1-hour charge, the uncooled single and dual packs heated at approximately 4.38 K/min and 2.71 K/min respectively, while the cooled packs rose by only about 0.09 K/min (single) and 0.07 K/min (dual), over a 97% reduction in rise rate once active cooling was engaged.

3.3 Comparison With and Without Active Cooling

Without cooling, the dual battery configuration demonstrated a marked increase in SoC without a substantial rise in temperature. The 69.3% improvement in SoC highlights the efficiency gains from the dual battery system,

while the 0.07% temperature increase shows that the thermal impact is minimal. With active cooling, the dual configuration maintained the same SoC advantage but showed an improved thermal response. The 0.45% decrease in temperature with cooling indicates that the dual battery system, combined with active cooling, offers a favorable balance between charging efficiency and thermal management. In essence, the simulation results suggest that the dual battery configuration enhances charging performance while maintaining manageable thermal levels, particularly when active cooling is used to regulate temperatures effectively. Figure 6 represents graphical illustration of vertex average of SoC against time for single and dual pack without active cooling. This indicates that dual pack vertex average of SoC in the absence of active cooling achieve slightly higher and steadier SoC retention with peak value of 0.551 compared to single pack with 0.325. This sustains better charging efficiency via steady sharing of current, while also minimizing wear rate, thereby, enhancing thermal management through the reduction of localized heat from high rate of discharge. It is observed in Figure 7, that single pack rapidly attended higher peaks as a result of concentrated heat, whereas, dual pack retains lower averages through improved dissipation, thereby, reducing the risk of thermal instability during charging while enhancing safety and efficiency. Hence, dual packs impact charging efficiency by minimizing thermal capacity loss, resulting in a comparatively lower peak temperature than the single pack under passive conditions. It should be emphasized, however, that neither configuration is adequately thermally managed without active cooling; both predicted peak battery temperatures (562.87 K for the single pack and 462.77 K for the dual pack) lie far above the normal safe operating range of Li-ion pouch cells (typically below 333–343 K) and enter a regime consistent with the onset of thermal runaway; the description of the dual-pack result as offering “passive thermal management” should therefore be read as a relative improvement over the single pack rather than as a demonstration of safe or satisfactory passive cooling.

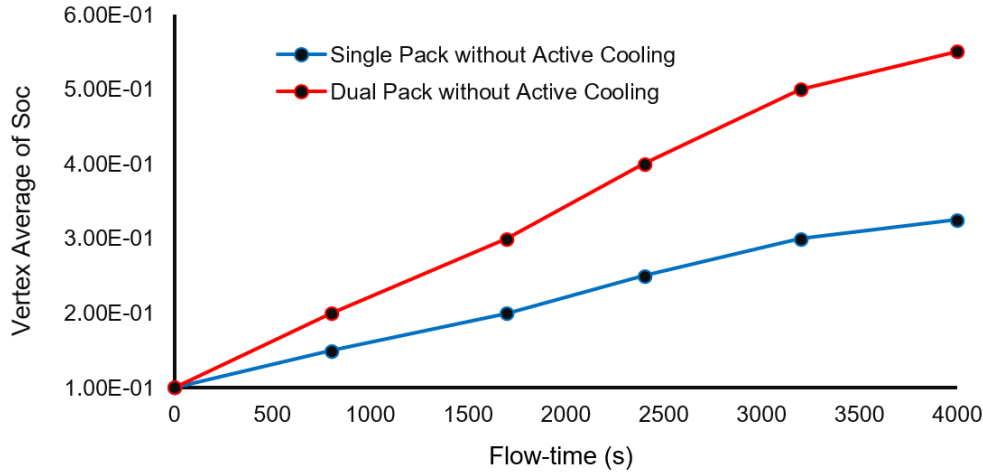


Figure 6. Vertex average of state of charge (SoC) against time for single and dual pack without active cooling

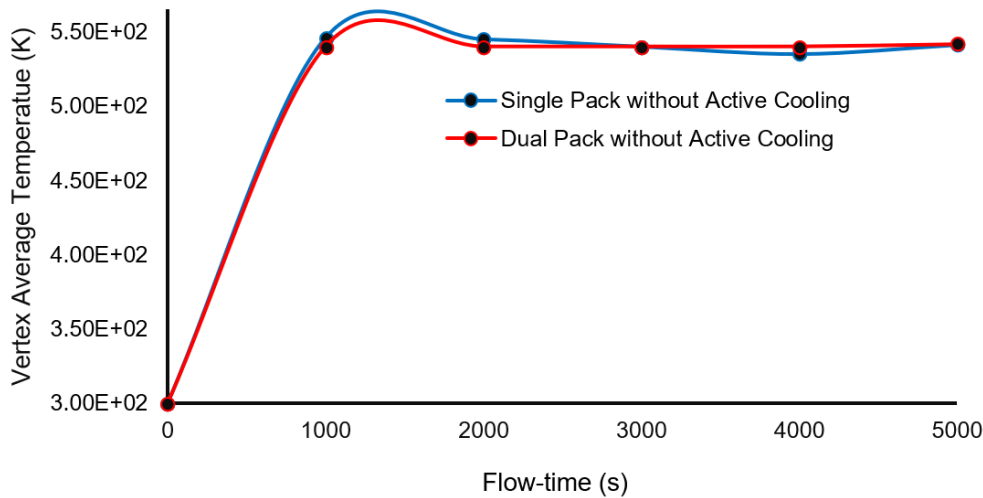


Figure 7. Vertex average temperature against time for single and dual pack without active cooling

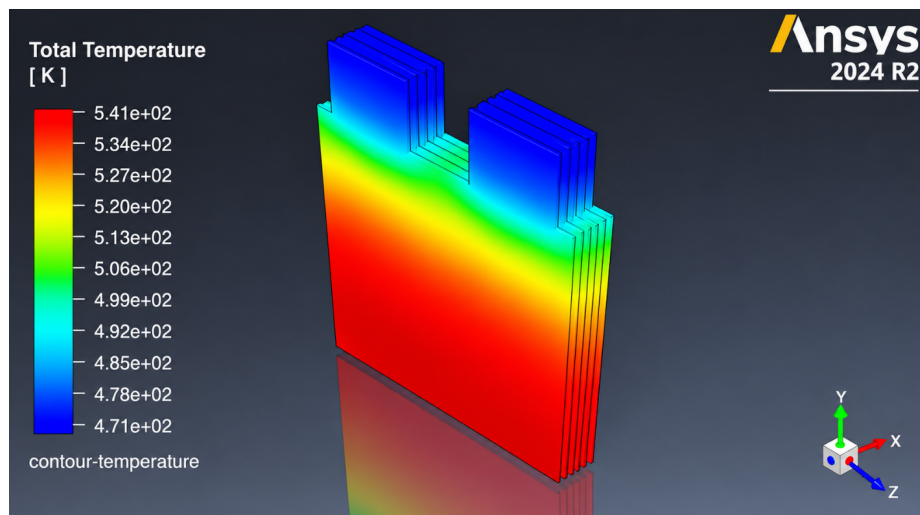


Figure 8. Contour of total temperature for single pack without active cooling

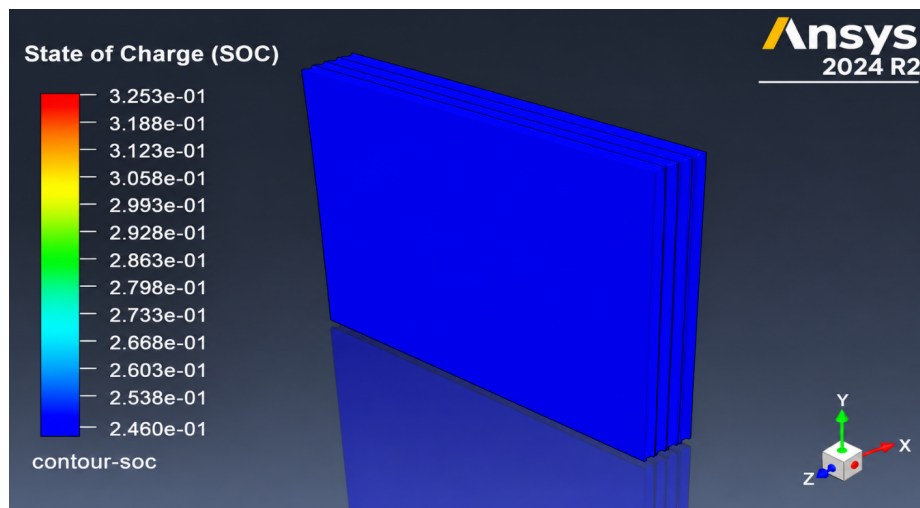


Figure 9. State of charge (SoC) contours for single pack without active cooling

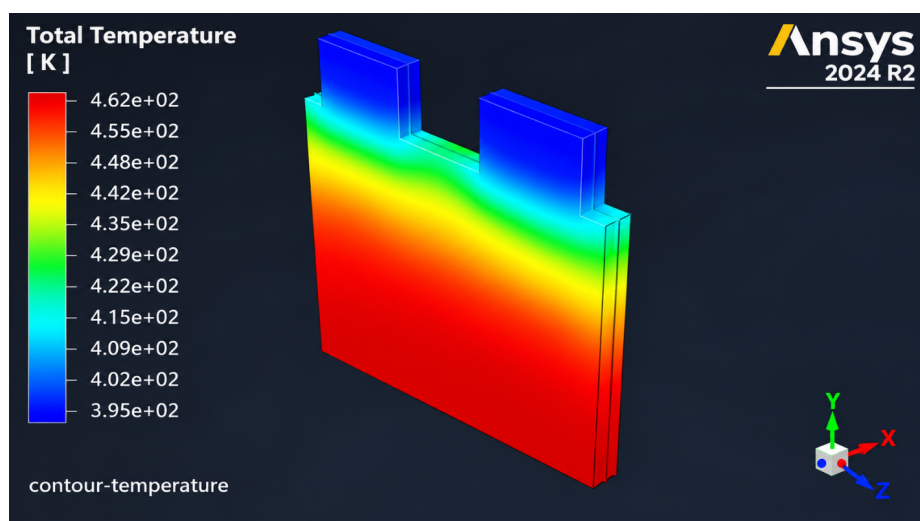


Figure 10. Contours of total temperature for dual pack without active cooling

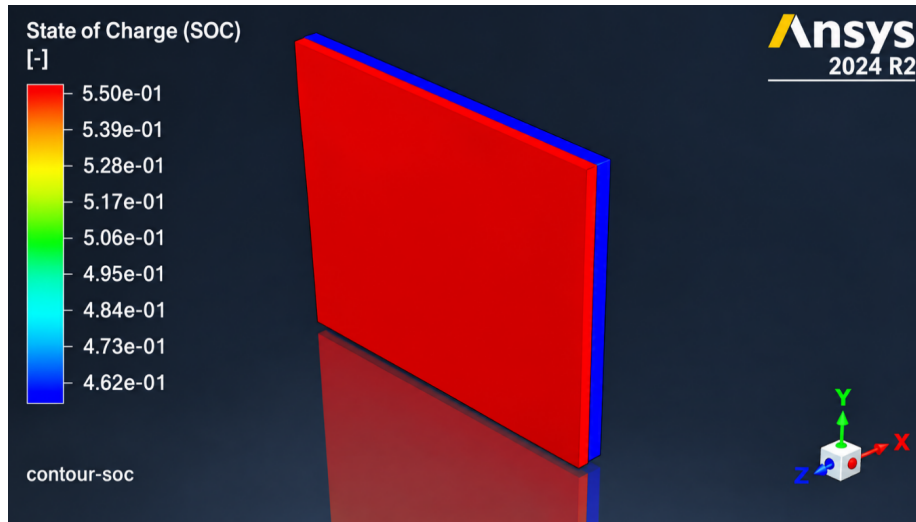


Figure 11. State of charge (SoC) contours for dual pack without active cooling

Total temperature contour in Figure 8 (without active cooling) illustrates single pack with intense hotspots reaching 541 K displayed by the red zones. As depicted in Figure 9, SoC contours without cooling display single pack's significant non-homogeneity (0.3253 (32.53%)), contrary to dual pack's proximately uniform high SoC domain. Dual pack battery (without active cooling in Figure 10) exhibits even distribution and slight cooler peak temperature reaching 462.77 K while SoC contours for dual pack without active cooling is presented in Figure 11. Therefore, dual pack manages to minimize temperature increase by sharing the thermal variation effects between the dual battery configurations, improving uniform efficiency. Hence, while single configuration exhibits an intense hotspot that severely compromises thermal performance, the dual configuration performs comparatively better without active cooling, though 541.11 K remains far in excess of a safe operating temperature and points to a real risk of thermal runaway rather than "thriving" passive thermal control; active cooling is therefore essential for either layout, not optional. Therefore, dual pack configuration facilitates equitable use, improving complete effectiveness while plummeting underutilized cells in EV functions.

Plot of SoC against time for single and dual pack with active cooling in Figure 12 demonstrates dual pack exhibiting higher SoC averages 0.5507 (55.07%) and 0.3253 (32.53%) pack configuration, with closer nature of the curves indicating optimum load sharing benefit for dual pack as well as efficiency under controlled settings. This lessens tem-SoC coupling impacts during charging. As observed in Figure 13, active cooling promotes cooler temperature for both configurations, but dual pack displays peak lowest average (291.77 K) compared to single pack's amplified residuals, significantly ameliorating charging speed, performance effectiveness and thermal management. Figures 14 to 17 which integrates active cooling indicate adequate SoC uniformity (>0.55) with nominal temperature gradient (250–300 K), surpassing single pack's remaining variations (SPRV). This is also referred to as "cell-to-cell" incongruity [33, 34], which relates to the variations in efficiency, voltage or capacity within each cell in a battery pack.

Active cooling scenarios in Figures 12–17 depicts cooling fluid dynamics which greatly favors dual battery pack configuration. Active cooling in this study is implemented as a liquid (water) coolant flow through a channel located along the base of the pack (Figure 2), as defined by the water properties in Table 3 and the coolant boundary conditions in Table 4, and this significantly enhances heat extraction compared to the passive (no-cooling) arrangement. In this study dual pack with active cooling is observed to attain the lowest vertex average temperatures (290–294 K) compared to single pack's higher residuals (290–296 K+). This demonstrates excellent convective heat transfer coefficient and uniform flow distribution around numerous packs, abating boundary layer resistance and hotspots. Therefore, temperature contours with minimal gradients is observed particularly in dual pack configuration, affirming that fluid dynamics efficiently balances thermal fields through adequately wetted surface area (WSA) and parallel flow paths (PFPs) [35, 36]. Therefore, SoC uniformity (>0.55 across dual pack) offers huge advantage, as stable lower temperatures decrease the effects of viscosity on coolant, while averting localized efficiency losses [37, 38]. Dual pack configuration enhances fluid flow dynamics (FFDs) by making provisions for interleaved or shared cooling channels, ameliorating Reynolds number-based turbulence for higher Nusselt numbers as well as the rates of heat removal [39, 40]. On the other hand, single pack configuration undergoes ineffective flow access and stagnation zones. Hence, dual pack integrated with active cooling should be prioritized for EV thermal management, enabling sustained high-efficiency charging and prolonged battery life. Regarding intra-pack spread, the cooled packs show

approximately modest 2.4–2.7 K difference between maximum and average temperature, consistent with the near-uniform contours in Figures 15 and 17, while the uncooled single pack shows a wider 21.75 K spread, reflecting the concentrated hotspot discussed in Section 3.3.

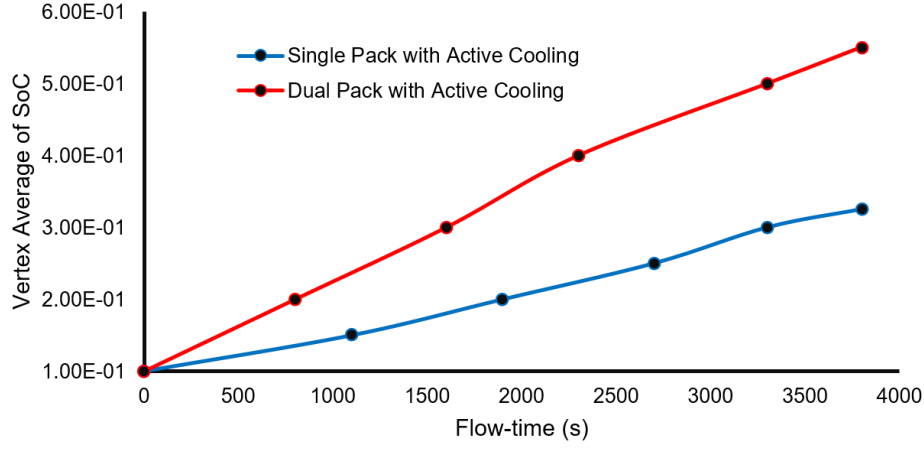


Figure 12. Vertex average of state of charge (SoC) against time for single and dual pack with active cooling

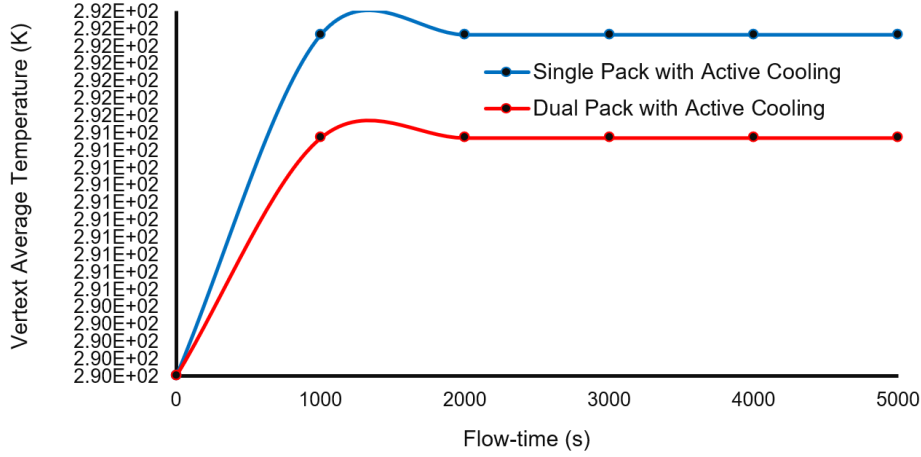


Figure 13. Vertex average temperature against time for single and dual pack with active cooling

3.4 Mathematical Validation

The SoC of a battery during charging can be expressed as:

$$SoC(t) = SoC_{\text{initial}} + \frac{I \cdot t}{C} \quad (1)$$

where, $SoC(t)$ is the SoC at time t , SoC_{initial} is the initial SoC, I denotes charging current (Amperes-A), t denotes time (seconds-s) and C is the battery capacity (ampere-hours-Ah). I is taken as the positive magnitude of the applied charging current ($I = 48$ A); the value of -48 A reported for “Specified System Current”/“battery-terminal-current” in Tables 2 and 5 reflects ANSYS Fluent’s internal sign convention (negative (-) denotes current flowing into the battery during charging) and is not used with its negative sign in the SoC equations. Since the unit of I is in A, t is in s, and C is in Ah, a conversion factor of 3600 (1 Ah = 3600 A·s) is required so that the current-time product is expressed in the same unit as capacity. Eq. (1) has therefore been amended to:

$$SoC(t) = SoC_{\text{initial}} + \frac{I \cdot t}{3600 \cdot C} \quad (2)$$

The corresponding change is carried through Eqs. (3)–(7). From Eqs. (3)–(5), $SoC_{\text{single}}(t) = \text{SoC}$ of the single-pack configuration at time t ; $SoC_{\text{dual}}(t) = \text{SoC}$ of the dual-pack configuration at time t ; $C_{\text{single}} = \text{nominal capacity of the single-pack configuration}$; $C_{\text{dual}} = \text{nominal capacity of the dual-pack configuration}$. For the single battery configuration:

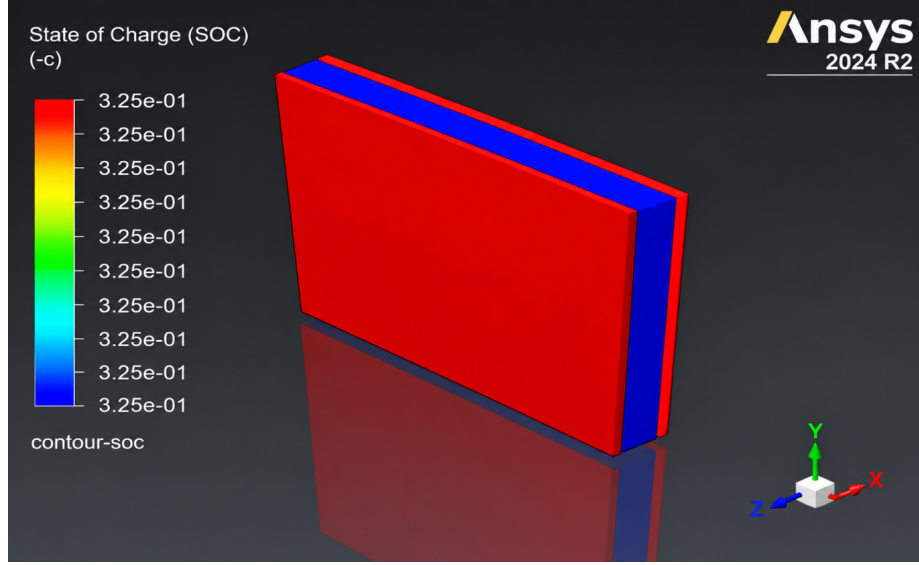


Figure 14. State of charge (SoC) contours for single pack with active cooling

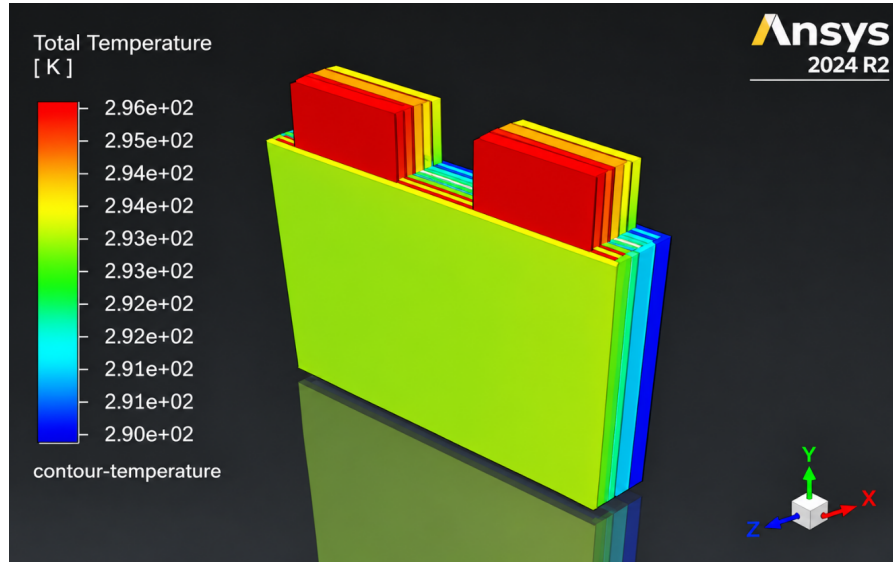


Figure 15. Contours of total temperature for single pack with active cooling

$$SoC_{\text{single}}(t) = SoC_{\text{initial}} + \frac{I \cdot t}{C_{\text{single}}} \quad (3)$$

For the dual battery configuration, where the total capacity is halved ($C_{\text{dual}} = 0.5C_{\text{single}}$):

$$SoC_{\text{dual}}(t) = SoC_{\text{initial}} + \frac{I \cdot t}{C_{\text{dual}}} = SoC_{\text{initial}} + \frac{I \cdot t}{0.5C_{\text{single}}} \quad (4)$$

This simplifies to:

$$SoC_{\text{dual}}(t) = SoC_{\text{initial}} + \frac{2 \cdot I \cdot t}{C_{\text{single}}} \quad (5)$$

The nominal/reference cell capacity configured in Fluent is 214 Ah for the single-pack model and 107 Ah for the dual-pack model, therefore, exactly half ($107 = 0.5 \times 214$). This directly supports the assumption $C_{\text{dual}} = 0.5C_{\text{single}}$ used in Eqs. (4) and (5). Hence,

$$\Delta SoC_{\text{single}} = \frac{I \cdot t}{C_{\text{single}}} \quad (6)$$

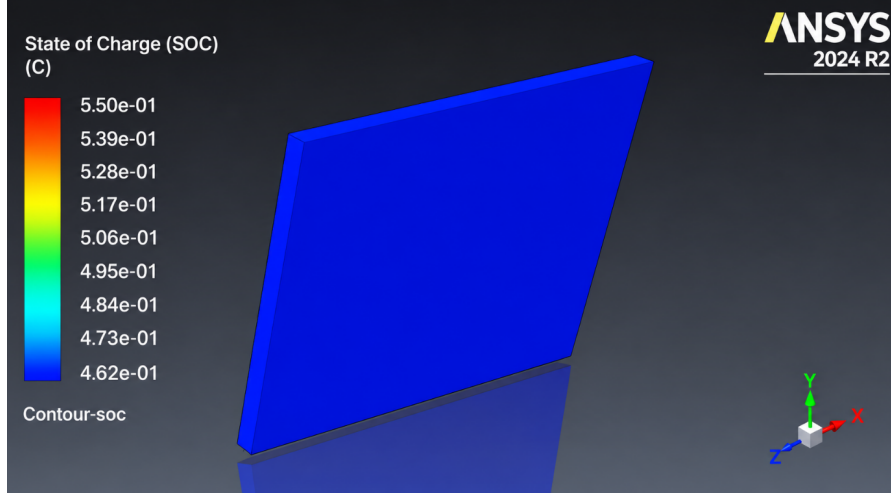


Figure 16. State of charge (SoC) contours for dual pack with active cooling

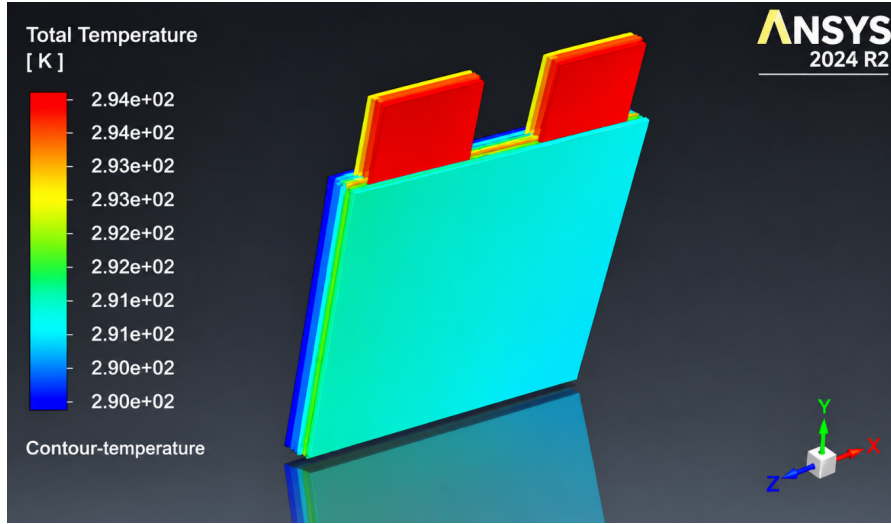


Figure 17. Contours of total temperature for dual pack with active cooling

While,

$$\Delta SoC_{\text{dual}} = \frac{2 \cdot I \cdot t}{C_{\text{single}}} \quad (7)$$

Therefore,

$$\Delta SoC_{\text{dual}} = 2 \cdot \Delta SoC_{\text{single}} \quad (8)$$

From Eqs. (6)–(8), $\Delta SoC_{\text{single}}$ = change in SoC of the single-pack configuration over the charging interval; ΔSoC_{dual} = change in SoC of the dual-pack configuration over the charging interval. This implies that dual battery configuration is supposed to attain twice the increase in SoC compared to the single battery configuration within the same charging time interval. However, this is not exactly the case owing to inadequacies arising from energy loss to heat. Therefore, exploring the integration of passive cooling techniques or more effective active cooling approaches could aid diminish the latent thermal challenges posed by dual battery configurations. This in turn can boost the effectiveness and usability of conventional EVs and hybrid-electric vehicles (HEVs) which depend highly on advanced Li-ion battery technologies to improve energy efficiency and thermal performance [41]. Battery-pack configuration and thermal management are critical to the safety, efficiency, and reliability of these systems. Modelling PLCs in single and dual battery packs, with and without active cooling, enables comparative assessment of thermal performance. Such analysis complements the broader sustainability objectives of HEV technologies by linking battery thermal management with efficiency, reliability, and environmental performance.

4 Conclusion

This study examined the effect of a dual battery configuration on enhancing the charging time of EVs, comparing it to a single battery pack in both cooled and non-cooled conditions. The simulation outcomes displayed a noteworthy improvement in the SoC for the dual battery configuration, which established a 69.3% increase in charging effectiveness compared to the single pack. Thermal performance was, however, the more decisive outcome of this study. Without active cooling, both configurations reached temperatures far beyond the safe operating range of Li-ion cells (541.11 K for the single pack and 462.77 K for the dual pack); the single-pack layout additionally displayed a pronounced hotspot and non-uniform temperature field, while distributing the same number of cells across two smaller dual packs lowered the peak temperature and improved uniformity, without eliminating the underlying risk. Active cooling resolved this decisively for both layouts: it lowered peak temperatures to near-ambient levels (295.62 K single, 294.18 K dual, making it total reductions of 47.5% and 36.4% respectively), reduced mean temperature by a further 0.45% in the dual pack relative to the single pack, and brought the intra-pack temperature spread down to about 2.4–2.7 K in both configurations, effectively removing the hotspot problem seen in the uncooled cases. These findings propose that espousing a dual battery configuration, provided it is paired with active cooling, could extensively lessen charging times while keeping the pack within a safe thermal envelope; without active cooling, neither configuration should be regarded as thermally acceptable for practical EV operation. However, some limitations were met during the investigation that could hamper its performance in actual scenarios. For example, owing to limited computational facilities, complete model of a outsized battery pack consisting of various cells could not be simulated. Rather, the study was constrained to modelling a miniature of pouch cells, which abridged the demonstration of the battery pack. This simplification possibly impacted the accurateness and scalability of the results, as battery packs in actual scenarios are much more intricate and would require thorough modelling of thousands of cells to capture realistic performances. Future work should extend this analysis to a full-scale, multi-cell pack and should systematically vary the coolant flow rate, channel arrangement (e.g., two-sided versus bottom or edge cooling, top-inlet/bottom-outlet versus bottom-inlet geometries) and ambient temperature, in order to establish how robust the thermal-management benefits of the dual-pack, actively-cooled configuration reported here are across the wider range of conditions relevant to power engineering and engineering thermophysics applications.

Author Contributions

Conceptualization, I.B.O. and I.I.E.; methodology, A.E.I. and O.D.E.; validation, O.D.E.; formal analysis, A.E.I. and O.D.E.; investigation, I.B.O.; resources, I.I.E.; data curation, I.B.O. and I.I.E.; writing-original draft preparation, A.E.I.; writing-review and editing, A.E.I.; visualization, O.D.E.; supervision, I.B.O. and A.E.I.; project administration, I.I.E. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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